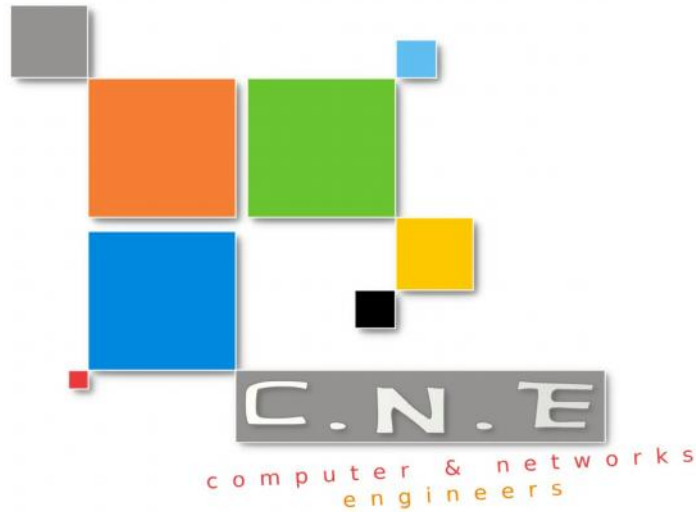


بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

اسئلة سنواته لهاده

البرمجه بلغه الكينونه



اتحاد طلبة هندسه الحاسوب والشبكات C.N.E

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Quiz 1 summer 2012, Object Oriented Programming
Name _____

Q1(10 points) For the following code answer the questions that follows

```
public class mainx {
    static int x=9;
    public static void main(String[] args) {
        int x=4;
        System.out.print(x);
        fun(x);
    }
    static void fun(int a)
    {
        System.out.print(x);
        System.out.print(fun((double)a));
    }
    static double fun(double g){
        return g+1.7;
    }
}
```

445.7

1- What is the output of this code when executed?

449.7

2- How many overloaded methods in this code?

2 over loaded method

3- Is variable g local to the method body for the method with signature fun(double g)

yes

4- Can we call none static method from a static method?

no

5- Which variable has the class scope?

x=a

6- How many copy constructors defined in the class mainx?

no copy constructor

Q2(5 points) given the following class definition answer the question that follows:

```
public class CatX {
    int x;
    float y;
    public CatX(){x=0; y=0;}
    public CatX(int a, float b){ x=a; y=b;}
    public void out(){ System.out.println((float) x+y); }
}
```

1- How many data members the class has?

2 D.M

2- What is the access modifier for the data members?

private

3- How many constructors we could define for this class?

5 constructor

4- What methods that has no return data type?

catX() {x=0; y=0;} out();

5- Which variable is casted from integer to float?

x

Q3(5 points)

Write a class with two doubles as private data members, name this class xxx, and your class should have two method one prints the sum of the data members and the other method finds the difference. Write instructions needed to instantiate an object of this class and call its methods.

public class xxx {

private double x, y;

public xxx() {};

public xxx(double a, double b) { x=a; y=b; }

public void out() { system.out.println(x+y); }

public void out() { system.out.println(x-y); }

public static void main (String args) {

double x=4;

double y=5.1;

double c, v;

xxx a=new xxx();

c = a.out(x+y);

v = a.out(x-y);

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Object Oriented Quiz

Name _____

Two pages

Q1(10 points) Given class A shown below, answer the questions that follows

```
public class A {
    public int x, y;
    public A(){ System.out.println("Default constructor call");}
    public A(int a, int b){ x=a;y=b; }
    public void read(int a, int b){ x=a; y=b; }
    public A add(A m){ x= x+m.x; y= y+m.y; return this;}
    public void out(){ System.out.print(x); System.out.print(y); }
    public void show(){System.out.println("Nothing");}
}
```

a- If the following instructions were executed in method main what will be the output?

```
A a1 = new A(); a1.read(1, 2); a1.out();
A a2= new A(4,5); A a3=a2.add(a1); a3.out();
```

1 2 ✓
5 7 ✓

Answer

b- Which object does the keyword *this* represent? Answer *current object*

c- Which method or function in the class takes an object in its parameter list?

add Answer *public A add(A m){ x= x+m.x; y= y+m.y; return this; }*

d- Which method accept object in its parameters?

e- Can we use the *this* keyword outside the class body? *No*

Q20 Given class B that inherits the class A in question, answer the question that follows?

```
public class B extends A {
    private double z;
    public B(int a, int b, double c){ z=c;}
    public void read(double a){ z=a;}
    public void print(){ this.out(); System.out.print(z);}
}
```

a- What will be the output if the following instructions were executed?

```
B b1=new B(1,2,3);
b1.print();
b1.read(1,2);
b1.read(5);
b1.show();
```

1 2 3.0
Nothing

Answer:

→ 1 2 3.0
→ the value of x = 1 y = 2
→ the value of z = 5
→ Nothing

b- How many data members in class B? *3 data members*

Q3(2 points) Answer the questions that follows for the class below

```
public class cascade {
    private int x;
    public cascade(int a){ x=a;}
    public cascade fun(){ System.out.print(++x); return this;}
}
```

a- What will be the instructions when this code is executed
a.fun().fun().fun().fun().fun();

A a=new A(0);

1 2 3 4 5

2

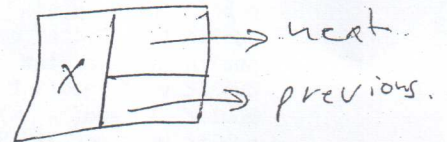
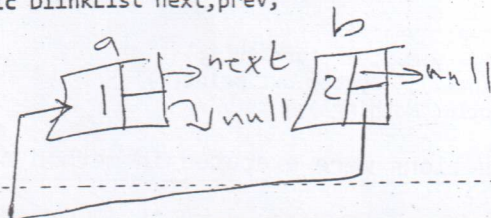
Object Oriented Quiz

Name _____

Two pages

Q4(3 points) write instructions to build two nodes of the double linked list class shown below.

```
public class DlinkList {
    public String str;
    public DlinkList next,prev;
}
```



a.x = 1

b.x = 2

a.next = b

a.previous = null

b.next = null

b.previous = a

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Q1(5 Points) Find the output of this code?

```
public class MyA {
    protected Integer x,y;
    public MyA(){System.out.println("Def MyA");}
    public MyA(Integer a, Integer b){ x=a; y=b;}
    public void values(){System.out.println(" MyA " + this.x + "," + this.y);}
    public void sum(){System.out.println(this.x+this.y);}
}

public class Mainx extends MyA {
    public Mainx(Integer a, Integer b){this.x=a; this.y=b;}
    public void values(){System.out.println("Mainx " + this.x + "," + this.y);}
    public static void main(String[] args) {
        MyA aref=new MyA(1,2);
        aref.values();
        aref.sum();
        Mainx bref= new Mainx(3,4);
        bref.sum();
        bref.values();
    }
}
```

Output 1

out put

MyA 1,2

3

MainX 3,4

x=1, y=2

x=3, y=4

Q2(5 Points)Find the output of this code

```
import java.io.FileReader;
public class Mainx extends MyA {
    public static void main(String[] args) {
        try{
            FileReader fr= new FileReader("Somestring");
        }catch(Exception e){
            System.out.println("Check file path");
        }finally{ System.out.println("Done"); }
        try{
            Integer x=1; x=0;
            System.out.println(1/x);
        }catch(Exception e){
            System.out.println("division error");
        }finally{System.out.println("complete");}
    }
}
```

Output 2

out put

Check file path

Done

division error

complete

Q3 (5 points) what this is the program output

```
public abstract class MyA {
    protected Integer x,y;
    public MyA(){System.out.println("Def MyA");}
    public MyA(Integer a, Integer b){ x=a; y=b;}
    public void values(){ System.out.println(" MyA " + this.x + "," + this.y); }
    abstract public void sum();
}
```

OOP Quiz, Name محمد بن عبد الله بن محمد الشعبة 200200

//Question 3 continues

```
public class Mainx extends MyA {
    public Mainx(Integer a, Integer b){ super(a,b);
        System.out.println("back from call");}
    public void sum() {System.out.println(this.x + this.y);}
    public void values(){super.values(); System.out.println("abc");}

    public static void main(String[] args) {
        Mainx ref1= new Mainx(1,2);
        ref1.sum();
        ref1.values();
    }
}
```

Output

Back from call
3
abc
MyA 1, 2

Call stack diagram showing the sequence of calls: MyA, then Mainx, and back to MyA.

Q4(5 points) Write a class so that the following instructions can be executed without errors.

```
A<Integer> ref1= new A<Integer>(1,2);
ref1.out();
ref1.add(4,5);
```

public class A<u>{

~~private int x, y;~~
public <u>X, <u>Y;

public void out() {System.out.println(x,y);};

public void add(int a, int b) {x=a, y=b};

}

public class Mainx {

A<integer> ref1= new A<integer>(1,2);

ref1.out();

ref1.add(4,5);

}